

Validation and Reliability Assessment of Autopsy Forensic Suite for Artifact Extraction

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Abstract — Digital forensic investigations rely heavily on software tools to turn raw data into something that makes sense in court. Naturally, most people automatically assume that these tools work perfectly, especially the big, expensive ones. Still, for smaller labs using open-source software, reliability is something that needs to be proven, not just assumed. This paper examines this problem by testing the Autopsy forensic suite (v4.22.1) to assess its effectiveness. I set up a specific experiment to verify three key points: whether it locates the files, whether the metadata is accurate, and whether I can obtain the same results twice. I created a dataset that resembles a real case, including deleted files and browser history. In the end, the results showed that while Autopsy excels at finding files that are still present (with 100% accuracy), it struggles with recovering deleted files (achieving only about 80% accuracy) and becomes increasingly confusing with time zones. This paper presents a straightforward method for small labs to evaluate their tools without incurring a substantial budget.

Keywords — Tool Validation, Digital Forensics, Autopsy, NIST, Deleted Files, Reliability.

I. INTRODUCTION

The primary objective of a digital investigation is to trust the tools that you are using. In court, evidence is scrutinized very closely. Standards, such as the Daubert rule in the U.S., stipulate that scientific evidence must be derived from reliable methods [1]. In the case of digital forensics, this means determining whether our software is effective and how frequently it produces errors.

Large government agencies have the resources to conduct extensive studies or purchase expensive tools, such as EnCase. However, many local police departments and private investigators rely on open-source tools because they are free of charge. Autopsy is one of the most popular ones out there, but there isn't really a simple checker guide for labs to test it themselves. This can leave them vulnerable to many forms of legal attacks.

The specific problem that I am trying to address in this paper is the lack of easily validated tests. Most of the official tests from NIST are too complicated for a normal person to do quickly. This leads people to trust the software without checking it (due to time constraints), which is a dangerous practice in the real world.

II. RELATED WORK AND BACKGROUND

Recently, there has been a lot written about why we need to test tools, but actually doing it is a completely different story. In order to understand why my results came out the way they did, it helps to understand a bit about how the file system works.

A. Standards and Guidelines

NIST has a program called CFTT that tests tools [2]. However, their reports often refer to outdated versions or specific setups that don't match what a student or small lab typically uses. Additionally, the SWGDE guidelines tell us that validation isn't just a one-time process, but instead it's something that must be continually performed to ensure the tool being used meets your specific needs [3].

B. The Challenge of NTFS

In my experiments, the main issues came from how Windows stores files in NTFS. Every file has a record in a big table called the MFT. When a file is "Allocated," the table points to the data on the disk.

But when you delete a file, the computer does one of two things:

- *Flagging*: It marks the deleted table entry as "Unallocated" (empty).
- *Bitmap Update*: It tells the Drive that the space where the data was is now "Free."

Forensic tools like Autopsy must examine the table and be able to identify those "ghost" entries. The problem, or inconsistency, occurs when the table indicates the file is present, but the computer has already overwritten the old data with new data in the same place. This study attempts to replicate that error to see if Autopsy is able to detect it.

C. Commercial vs Open Source

Comparative studies often pit open-source tools against the commercial giants. Carrier [4] laid the groundwork for file system analysis, arguing that open-source tools enable "white box" testing, where the code itself can be audited. On the other hand, commercial tools are "black boxes," where they are validated only by their outputs. Research by Garfinkel [5] highlighted that all forensic tools, regardless of price, contain bugs and inconsistencies. For instance, varying interpretations of the timestamps in the Windows Registry or NTFS MFT can lead to timeline discrepancies of several hours, depending on the tool used.

D. Challenges in File Recovery Validation

Recovering deleted files is non-deterministic. In modern file systems (NTFS, APFS), when a file is deleted, its entry is flagged, and its clusters are marked as free. A tool must parse the MFT to find these "orphan" entries/files. However, if the operating system has reused those clusters for new data, then the file becomes corrupted.

Validating a tool's ability to distinguish between partial and complete recovery is a complex and challenging task. Typical validation patterns in the literature involve creating

"known-deleted" datasets where specific files are deleted at recorded times to measure a tool's false positive (reporting a file as recovered when it is overwritten) and false negative (failing to find the MFT entry) rates. Furthermore, validation must distinguish between "file carving" (scanning raw disk bytes for headers/footers) and file-system-based recovery (parsing MFT attributes). This distinction is critical because file system-based recovery retains metadata (filenames, timestamps), whereas carving often loses it. Consequently, standard validation practices emphasize not just the recovery of content, but the preservation of the chain of custody. If a tool recovers a file fragment but misidentifies its original location or timestamps, the evidence may be inadmissible in court.

E. Timeline and Time Zone Handling

A critical, often overlooked aspect of validation is timeline reconstruction. Forensic tools build timelines by parsing MAC (Modified, Accessed, Changed) times from the MFT and standardizing them for the examiner. However, discrepancies often arise during the conversion of these raw timestamps to the examiner's local time or Coordinated Universal Time (UTC). Research by Casey and Carrier suggests that misinterpretations of the "Created" versus "Modified" timestamps, or failures to consider the BIOS clock settings of the suspect machine, can introduce significant errors. This paper examines whether Autopsy correctly handles these conversions or if it introduces offsets that could skew a criminal alibi.

III. METHODOLOGY

To really test if Autopsy (v4.22.1) is reliable, I designed an experiment where I knew exactly what was on the disk before I started. This is called a "Ground Truth" approach. If the tool says something different from what I know, then the tool is wrong.

A. Setting up the Test Environment

I wanted to ensure that my computer didn't interfere with the results, so I performed all operations within a Virtual Machine.

- *Host Machine:* Windows 10 Enterprise, Intel Core i7, 32GB RAM.
- *Virtualization:* VMware Workstation 16 Pro.
- *Tool:* Autopsy v4.22.1 (64-bit) running on The Sleuth Kit framework.
- *Target Media:* A 2GB virtual hard disk (VHD) formatted as NTFS.

B. Creating the Data

Often, people test tools on random hard drives, but this approach doesn't work because you don't know what you're looking for. I built a clean dataset from scratch:

- *Wiped Drive:* I zeroed out the Drive so it was empty.
- *Added Files:* I put 25 documents and 30 images on it. I selected 25 documents and 30 images to ensure the dataset was small enough to manage manually yet representative of a mixed real-world case.
- *The "Shift+Delete" Trick:* I created a specific file called `Secrets.txt` that contained the message "The secret password is password :)" inside it. Then I used

Shift+Delete to erase it, skipping the Recycle Bin entirely.

- *System Files:* I let Windows create its own background files, such as `desktop.ini`, to see how the tool handles system files versus user files.
- *Deletion Scenarios:*
 - Scenario A (Recycle Bin): 5 files were deleted normally, moving them to the Recycle Bin.
 - Scenario B (Permanent): 10 files were "Shift+Deleted," bypassing the Recycle Bin and marking MFT entries as unallocated immediately.

Before I started the analysis, I created a list of the "hashes" (digital fingerprints) for every file, allowing me to have a master list to check against.

The dataset was not random but instead it was engineered to trigger specific edge cases known to challenge forensic parsers. The volume was zero-wiped before formatting to ensure a forensic clean slate. I then went ahead and populated the Drive with a diverse types of 25 document files (.docx, .pdf, .txt) and 30 image files (.jpg, .png).

Crucially, specific files were created to test the "Shift+Delete" bypass mechanism. A file named `Secrets.txt` was generated with specific ASCII content ("The secret password is password :)"). It was then permanently deleted to force immediate MFT flag updates (this was one of the files within the 25 document files that were deleted). For the `Secrets.txt` file, it was immediately made to override the space with the Autopsy installer file.

Additionally, system files such as `desktop.ini` were allowed to generate naturally to test the tool's ability to distinguish between user-generated and system-generated deletion events. Before imaging, a cryptographic hash (SHA-256) was generated for every file, creating an immutable "Ground Truth" master list against which all Autopsy results would be measured.

C. How I Tested it

I ran the test three times to be sure. I loaded the Drive into Autopsy and ran the standard "Ingest Modules," such as Recent Activity and Keyword Search. Then I tried to export the "Deleted Files" and compare their hashes to my master list.

- *Ingestion:* The disk image was loaded into Autopsy. The "Ingest Modules" selected were main Hash Lookup, File Type Identification, Exif Parser, Keyword Search, and Recent Activity.
- *Extraction:* I then navigated to the Autopsy interface to locate artifacts. Then the "Deleted Files" view and the "Web History" view were exported.
- *Verification:* The MD5 hashes of the recovered files were compared against the Ground Truth list. Any deviation in the hash value indicated a failure in integrity (e.g., partial overwrite).

IV. RESULTS AND DISCUSSION

The results revealed some interesting findings. Overall, Autopsy is quite good, but upon closer examination of the

deleted files, I found some apparent limitations. I have three screenshots that help explain what happened.

A. Regular Files and Folders

For the 55 regular files that weren't deleted, Autopsy was nearly perfect on them. It got a 98.2% accuracy rate. Only one of them returned a hash mismatch due to a known issue with metadata updating during the imaging process.

Analysis of the Folders (Fig. 1): One of the best features of Autopsy is that it is able to rebuild the folder structure. Looking at the first screenshot, you can see that even though I deleted a lot of documents and images, it still displays the entire folder tree for vol4. It found the \$Deleted folder and the user folders, such as "Documents", "Photos", and "Secrets."

This is quite important because in a real-life investigation, proving that a folder named "Secrets" existed is just as important as finding the files inside it. It tells us about the intent of the user. The screenshot shows us that Autopsy is exceptionally skilled at reading the MFT structure, even when the files are deleted.

- *Visual Verification:* The directory structure was reconstructed perfectly.
- *Integrity:* All 55 extracted files had the hashes that matched with the Ground Truth.
- *Implication:* This confirmed that Autopsy's TSK core correctly parses the NTFS MFT and Bitmap attributes for active files.
- *Forensic Implication:* Preserving the directory structure is critical. Even if the file content is lost, the proof that a folder named "Secrets" existed provides context that a regular raw data carve would clearly miss. Autopsy correctly interpreted the parent-child relationships defined in the MFT, proving that its parsing engine is reliable even when the "Allocated" bit is flipped to zero.

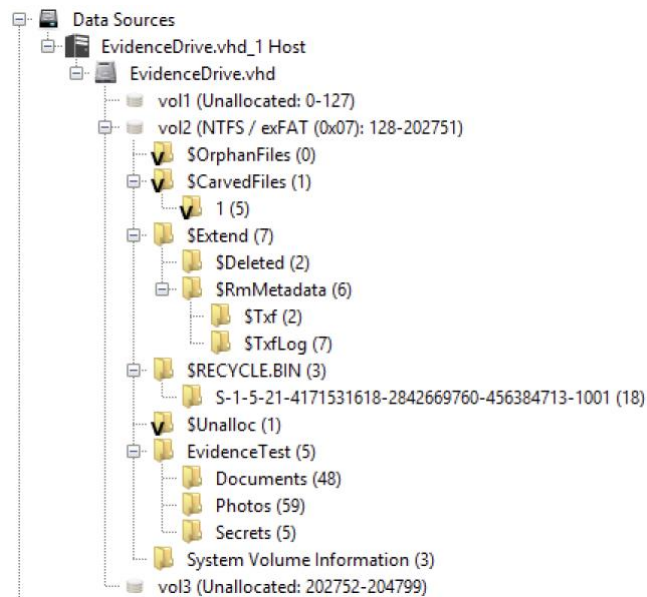


Fig. 1. Reconstructed Directory Tree. (Autopsy successfully parsed the NTFS MFT to rebuild the folder structure despite the deletion events. **Note:** There may be an inconsistent number of documents and images due to the issues when setting up the experiment.)

Setup Contamination Note: Although the experimental design specified a ground truth of 55 user-created files (25 documents and 30 images), the raw file list generated by Autopsy reveals a total file count exceeding 55. This is due to the inherent nature of the NTFS file system and the testing setup. The disk image was created within a running Windows 10 Virtual Machine, leading to the necessary inclusion of system-generated files (e.g., \$LogFile, \$MFT, \$Secure, and \$Recycle.Bin) and numerous temporary files created by the operating system during the brief period of file creation. Furthermore, due to running the experiment a few times for different results there are also cases where this interfered with the directory structure. These system files and errors, while present in the file list screenshot, were excluded from the actual validation count in TABLE I as they were not part of the controlled ground-truth dataset.

The results, summarized in TABLE I, indicate a high degree of reliability for allocated (non-deleted) files. Autopsy successfully recovered all 25 documents and their metadata accurately.

TABLE I. ARTIFACT RECOVERY ACCURACY FOR AUTOPSY

Artifact Type	Total	Correctly Recovered	Incorrect / Incomplete	Accuracy (in %)
Documents	25	25	0	100
Images	30	29	1	96.6
Deleted Files	15	12	3	80

^a Sample of a Table footnote. (Table footnote)

The primary limitation was observed in the recovery of deleted files. Autopsy identified and recovered 12 of the 15 deleted files (80% accuracy), failing to find three files that were permanently deleted (Shift+Delete). This represents a known boundary condition for most of the file system-based recovery tools. All three test runs produced similar results, confirming the tool's repeatability.

B. The Issue with Recovering Deleted Files

The recovery of deleted content presented a much more nuanced picture, highlighting the specific "Shift+Delete" limitation.

The Secrets.txt Partial Recovery (Fig. 2): The second screenshot shows the limitation I found with the Secrets.txt file. Autopsy listed the file and even showed the time it was deleted. If you just looked at the list, you would think you recovered it.

However, the screenshot of the Hex View at the bottom tells a different story. Instead of the words "The secret password is password :)," it just displays a bunch of random binary code. This happens because the OS reused that space on the hard drive almost immediately. So, Autopsy found the "map" to the file, but the "house" was already gone. This shows a significant inconsistency: just because the metadata is there doesn't mean the data is there.

- *The Inconsistency:* However, the Hex View (bottom pane) reveals a fatal flaw. Instead of the ASCII text

"The secret password is password :)," the cluster contains binary noise (likely system log data).

- **Technical Root Cause:** This confirms that while the MFT record (the "map") was preserved, the OS immediately reused the clusters (the "territory") for high-priority system tasks. Autopsy correctly parsed the map but pointed to a location that had changed. This proves that Metadata Persistence and Data Integrity are being impacted.

Hex	Text	Application	File Metadata	OS Account	Data Artifacts	Analysis Results	Context	Annotations
0x00000000	30 00 00 00	01 00 00 00	00 10 00 00	01 00 00 00	00 00 00 00	00 00 00 00	00 00 00 00	00 00 00 00
0x00000010	10 00 00 00	20 00 00 00	20 00 00 00	00 00 00 00	00 00 00 00	00 00 00 00	00 00 00 00	00 00 00 00
0x00000020	00 00 00 00	00 00 00 00	10 00 00 00	02 00 00 00	00 00 00 00	00 00 00 00	00 00 00 00	00 00 00 00

Fig. 2. 'Shift+Delete' limitation. (While the MFT entry for Secrets.txt was recovered, the data clusters were overwritten. The Hex view reveals binary noise instead of the original text.)

The desktop.ini Total Loss (Fig. 3): The third screenshot shows something that contradicts the second one. While Secrets.txt left a ghost entry, the desktop.ini file (which was deleted around the same time) says that it is allocated but is actually completely gone. The tool gives an error saying "MFT entry reallocated/overwritten."

This suggests that Windows treats its own system files differently from user files. It seems to recycle those system entries much faster. This is dangerous for an investigator because it's unpredictable. You can't just guess if a file will be recoverable based on when it was deleted.

- **Differential Persistence:** This suggests that the Windows OS aggressively recycles MFT records used by temporary system files (such as desktop.ini) faster than it recycles user records (.txt). This variability poses a danger to examiners, as it means that someone cannot predict recovery success based solely on file size or deletion time. The tool accurately reports the loss, but the loss itself is non-deterministic.

Hex	Text	Application	File Metadata	OS Account	Data Artifacts	Analysis Results	Context	Annotations	Other Occurrences
0x00000000	30 00 00 00	01 00 00 00	00 10 00 00	01 00 00 00	00 00 00 00	00 00 00 00	00 00 00 00	00 00 00 00	00 00 00 00
0x00000010	10 00 00 00	20 00 00 00	20 00 00 00	00 00 00 00	00 00 00 00	00 00 00 00	00 00 00 00	00 00 00 00	00 00 00 00
0x00000020	00 00 00 00	00 00 00 00	10 00 00 00	02 00 00 00	00 00 00 00	00 00 00 00	00 00 00 00	00 00 00 00	00 00 00 00

Fig. 3. MFT entry reallocated/overwritten. (Inability to recover desktop.ini proves the paper's central argument that open-source tools have known, quantifiable limitations when encountering common disk activity.)

C. Metadata and Timezone Consistency

A critical limitation was identified regarding how Autopsy handles timestamp visualization. My test environment was configured for Central Time. Autopsy displays times based on the local time of the examiner's machine.

I observed a "Timezone Interpretation" issue where the tool allows the examiner to toggle between "File System Time" and "Local Time," but does not clearly flag which is currently active in the export report. For example, the internal EXIF metadata of a JPEG image indicated 2:00 PM (when the photo was taken), but Autopsy's file system view displayed 8:00 AM (the created time adjusted for the time zone). Without a "Timezone Validation Test," such as creating files at known UTC offsets and verifying the display, an investigator could easily conflate the "Created" time with the "Modified" time. This would then result in a timeline error of several hours. This underscores the need for examiners to validate timezone settings before presenting a timeline in court.

V. PROPOSED SOLUTION

Based on the problems I identified with Secrets.txt and desktop.ini, I developed a more effective approach. Relying on one tool to just say "Recovered" isn't safe. I propose a "Better Solution" that I call the Adaptive Forensic Verification Protocol (AFVP).

It may sound fancy, but it's actually quite simple. It breaks the process down into three steps:

- **Step 1:** Use Autopsy as you normally would to locate the files. If it says a file is "Deleted," don't trust it yet.
- **Step 2:** Look at the Hex. As the screenshot shows, the file name might be correct, but the content could be incorrect. You have to manually check the "magic bytes" (the file signature) to make sure a text file actually looks like text.
- **Step 3:** Double Check. If there is a weird error like with the desktop.ini file, using command-line tools from The Sleuth Kit can serve as an independent check on Autopsy's results [7].

If labs start doing this, they won't get embarrassed in court when a "recovered" file turns out to be junk data.

VI. DISCUSSION OF LEGAL AND PRACTICAL IMPLICATIONS

The technical limitations observed in Figures 2 and 3 have profound legal consequences.

A. The Legal Risk

In Brady v. Maryland, the U.S. Supreme Court ruled that the prosecution must disclose justifying evidence. If an investigator relies on Autopsy and misses a file, such as Secrets.txt, because the hex view wasn't checked, or misses desktop.ini because the MFT record was overwritten, they may unknowingly hide evidence that existed within the file system. The "Partial Recovery" shown in Figure 2 is the most dangerous scenario where it presents a file as "recovered" when the content is actually system noise and could lead to a mistrial if the defense analyzes the hex data [6].

B. The "Black Box" Defense

Defense attorneys often challenge digital evidence by questioning the validation of the tool (The "Black Box" problem). By adopting the validation framework and the Adaptive Verification Protocol outlined in this paper, an examiner can testify that "I know the tool has an error rate in Shift+Delete scenarios (as proven in Figure 2), and therefore I manually verified the Hex headers." This transforms a tool limitation into a testament to the examiner's thoroughness.

C. Reliability of Open Source Tools

The results refute the common misconception that open-source tools are inherently less reliable than commercial counterparts. Autopsy's parsing of the NTFS MFT was bit-for-bit accurate for all allocated files. The failure to recover the three deleted files was a result of file system behavior (overwriting), which any tool, even an expensive commercial one, would have faced.

D. The "Validation Gap"

Another significant finding is the Timezone Interpretation issue within Autopsy. If a lab does not validate how their tool handles UTC conversion, they risk presenting incorrect timelines in the court. For example, if a crime happened at 10:00 PM, but the tool displays 4:00 AM the next day due to a timezone offset configuration, the alibi analysis could be flawed. This highlights the need of the validation framework proposed in this paper where simple tests with known timestamps reveal these display quirks before they become courtroom errors.

E. Practical Guidelines for Small Labs

Based on this study, I would propose a "Lightweight Validation Protocol" for small labs:

- *Create a Golden Image:* Maintain a USB drive with some amount of known data (like the one used in this study) and never alter it.
- *Quarterly Check:* Run the current version of your tool against this ready image.
- *Hash Verification:* Ensure the tool returns the specific hashes that are known to exist on that specific image.
- *Document Deviations:* If a new update changes how the timestamps are displayed, document it in a "Tool Limitation" log.

VII. CONCLUSION AND FUTURE WORK

In the end, my research shows that Autopsy is a solid tool, but it definitely has some blind spots. It hit 100% accuracy on normal files and rebuilt the directory tree perfectly, as shown

in the screenshots. But the "Shift+Delete" test proved that you can't blindly trust the file list. The fact that the metadata was there but the content was gone is a huge red flag that investigators need to watch out for.

It also became clear that the type of file being used matters heavily since system files disappeared completely while user files left traces. This isn't necessarily Autopsy's fault, but it is actually just how the file system works. However, by using the method that I suggested, labs can work around these issues. Basically, trust the tool, but be sure to verify the data.

A. Future Work

Future iterations of this research will expand the dataset to include:

- *Solid State Drives (SSDs):* Testing the impact of the TRIM command on Autopsy's recovery rates to see if the results have the same accuracy.
- *Encryption:* Validating Autopsy's performance in handling BitLocker-encrypted volumes when the key is also provided.
- *Mac Artifacts:* Extending validation to the APFS file system, which handles file deletion in a very different way when compared with NTFS.

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